Date	28 October 2026	Team ID	EN25CS3011166
Project Name	Smart Healthcare & Cardiac Risk Analytics	Maximum Marks	5 Marks

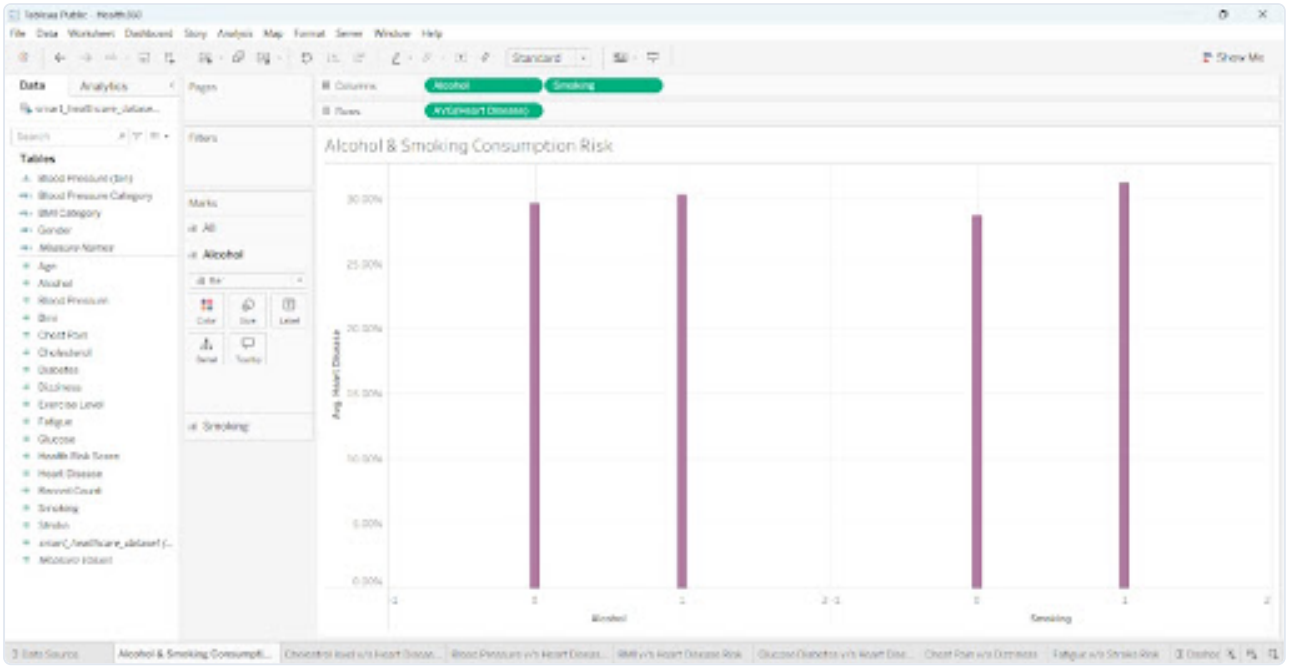
BUSINESS QUESTION AND VISUALIZATION REPORT

VISUALIZATION DEVELOPMENT & SNOWFLAKE ANALYTICS

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex health and medical datasets into visual formats that are easy to interpret, enabling clinicians and analysts to gain insights and make informed clinical decisions.

Q1. How do Alcohol and Smoking consumption patterns impact the Average Heart Disease risk?

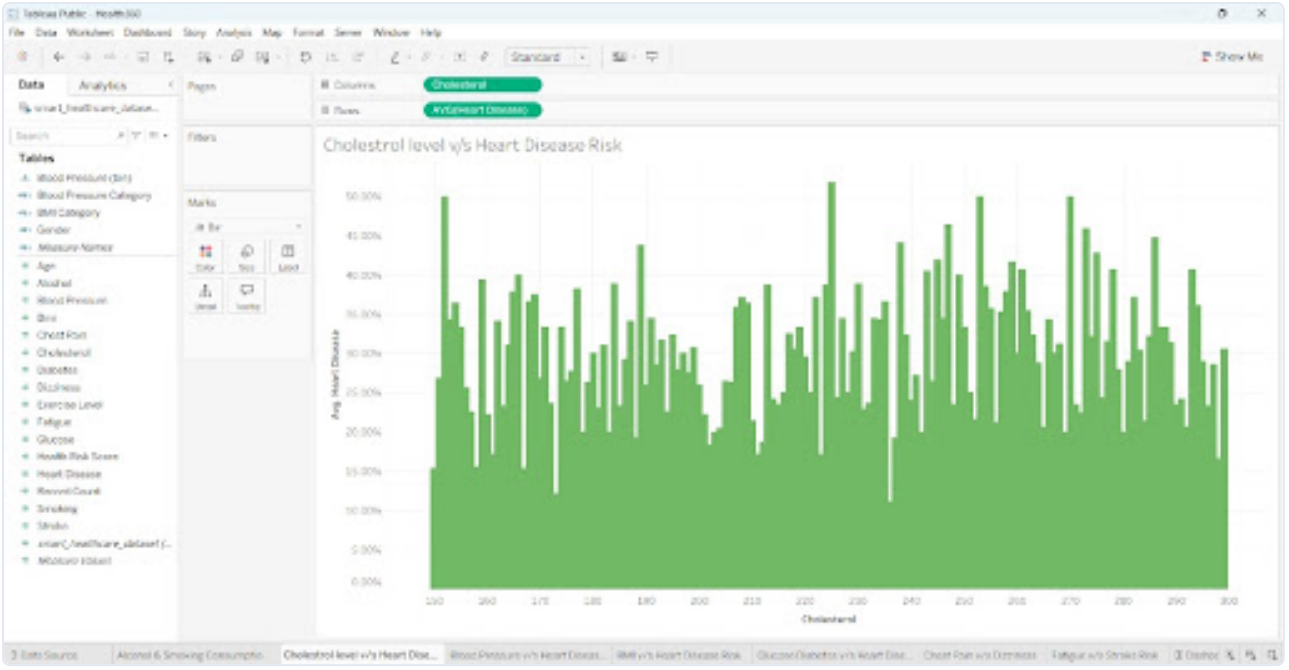
```
-- Snowflake SQL Query SELECT ALCOHOL, SMOKING, AVG(HEART_DISEASE) AS AVG_HEART_DISEASE F
ROM SMART_HEALTHCARE_DATASET GROUP BY ALCOHOL, SMOKING ORDER BY ALCOHOL, SMOKING;
```



Visualization 1: Alcohol & Smoking Consumption Risk (Bar Chart)

Q2. What is the relationship between Cholesterol levels and Heart Disease Risk?

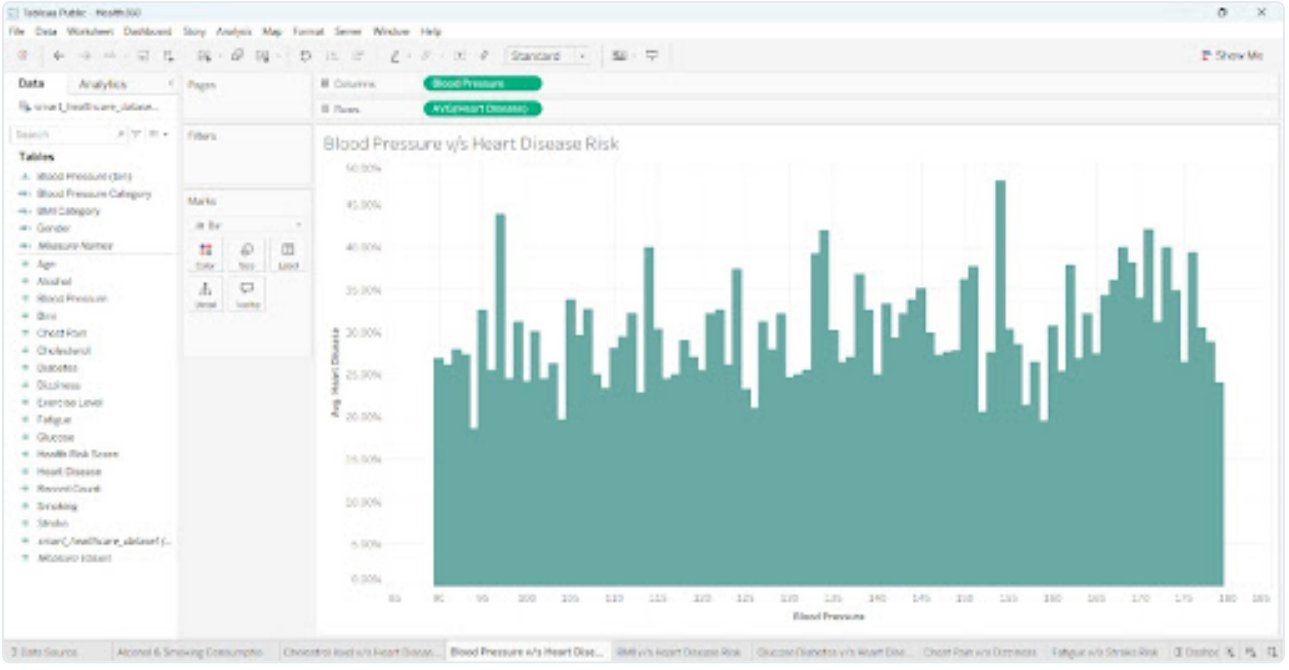
```
-- Snowflake SQL Query SELECT CHOLESTEROL, AVG(HEART_DISEASE) * 100 AS AVG_HEART_DISEASE_
PCT FROM SMART_HEALTHCARE_DATASET GROUP BY CHOLESTEROL ORDER BY CHOLESTEROL ASC;
```



Visualization 2: Cholesterol level v/s Heart Disease Risk

Q3. How does variation in Blood Pressure correlate with Heart Disease Risk?

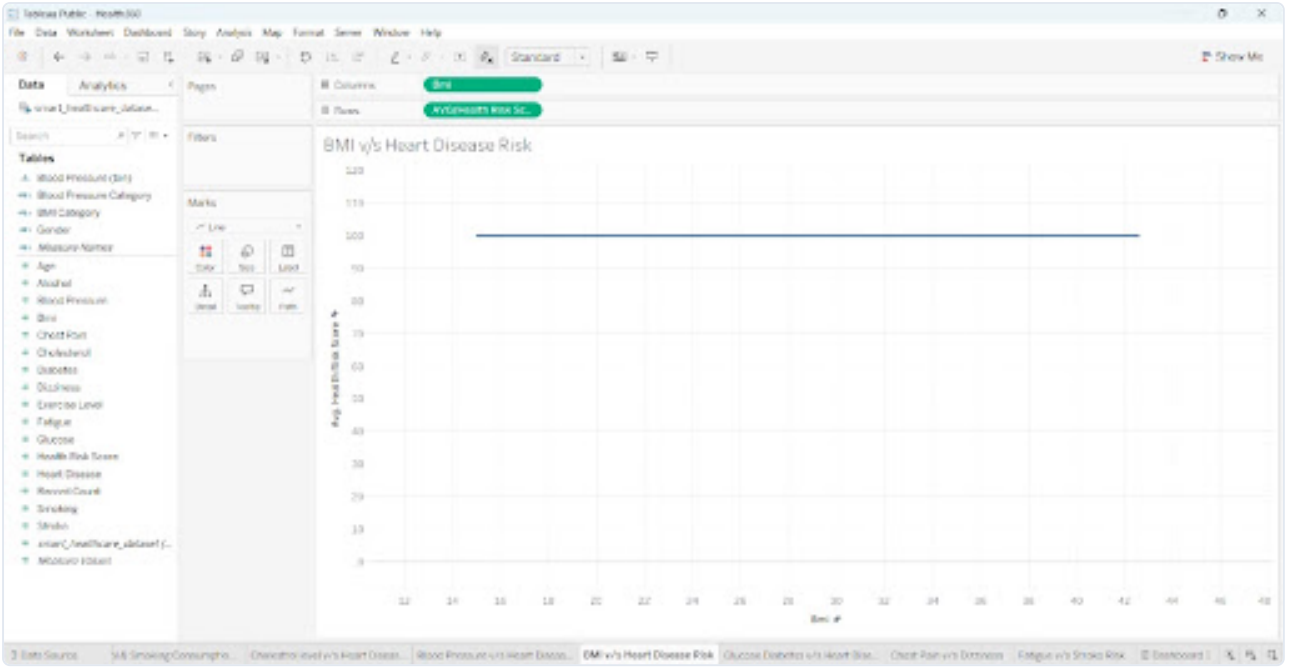
```
-- Snowflake SQL Query SELECT BLOOD_PRESSURE, AVG(HEART_DISEASE) AS AVG_HEART_DISEASE FRO
M SMART_HEALTHCARE_DATASET GROUP BY BLOOD_PRESSURE ORDER BY BLOOD_PRESSURE ASC;
```



Visualization 3: Blood Pressure v/s Heart Disease Risk

Q4. What is the trend of overall Health Risk Score across different BMI levels?

```
-- Snowflake SQL Query SELECT BMI, AVG(HEALTH_RISK_SCORE) AS AVG_HEALTH_RISK_SCORE FROM S
MART_HEALTHCARE_DATASET GROUP BY BMI ORDER BY BMI ASC;
```



Visualization 4: BMI v/s Heart Disease Risk Line Trend

Q5. How do Glucose and Diabetes indicator levels affect Heart Disease Risk?

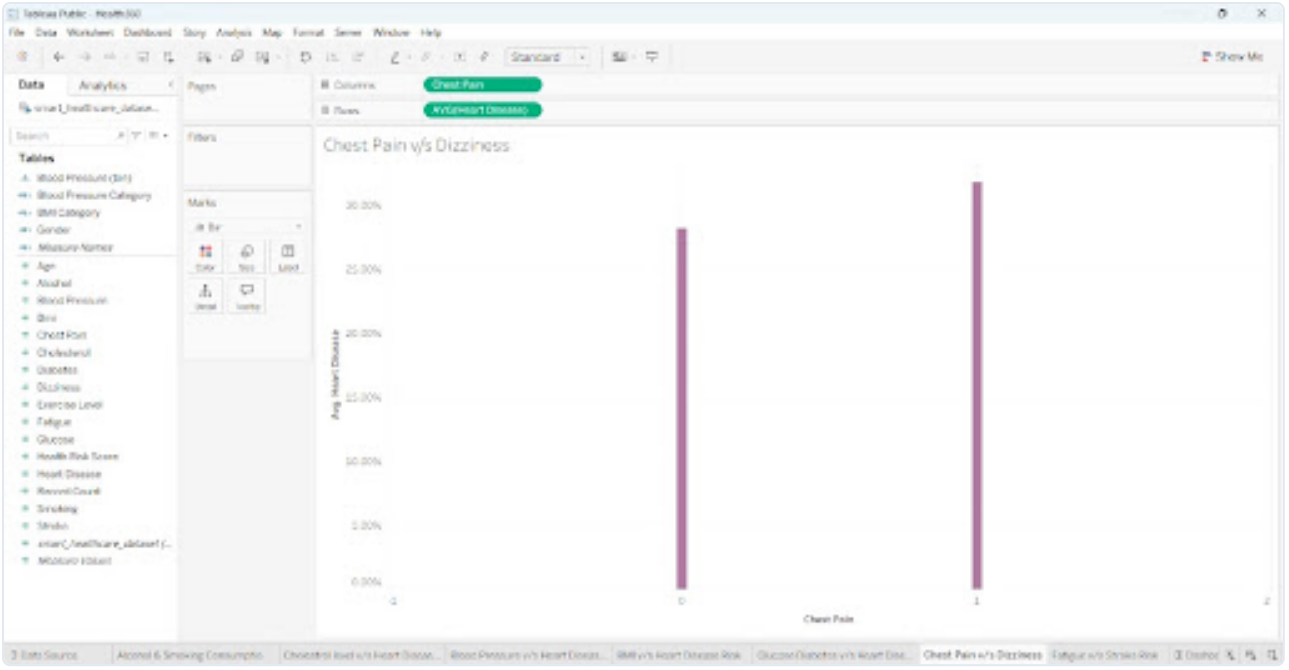
```
-- Snowflake SQL Query SELECT GLUCOSE, DIABETES, AVG(HEART_DISEASE) AS AVG_HEART_DISEASE
FROM SMART_HEALTHCARE_DATASET GROUP BY GLUCOSE, DIABETES ORDER BY GLUCOSE ASC;
```



Visualization 5: Glucose Diabetes v/s Heart Disease Risk

Q6. Does the presence of Chest Pain correlate with increased instances of Dizziness?

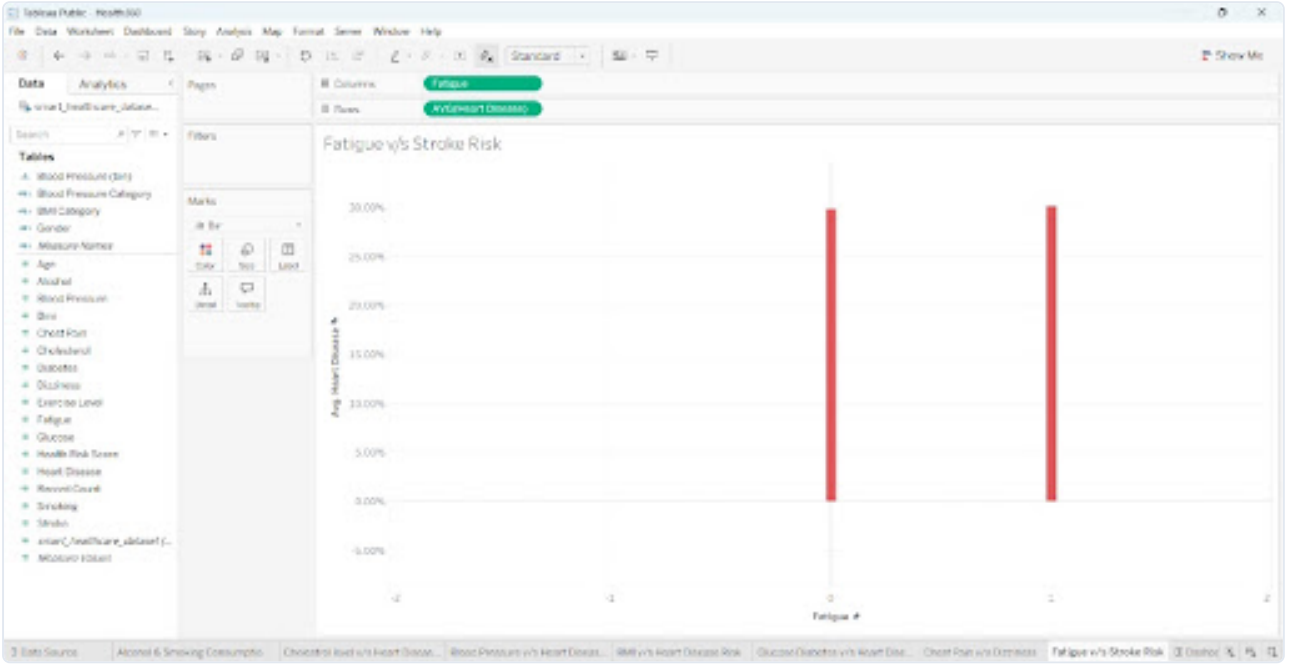
```
-- Snowflake SQL Query SELECT CHEST_PAIN, AVG(HEART_DISEASE) AS AVG_HEART_DISEASE FROM SM
ART_HEALTHCARE_DATASET GROUP BY CHEST_PAIN;
```



Visualization 6: Chest Pain v/s Dizziness & Heart Risk

Q7. How does Fatigue status impact overall Stroke Risk among patient groups?

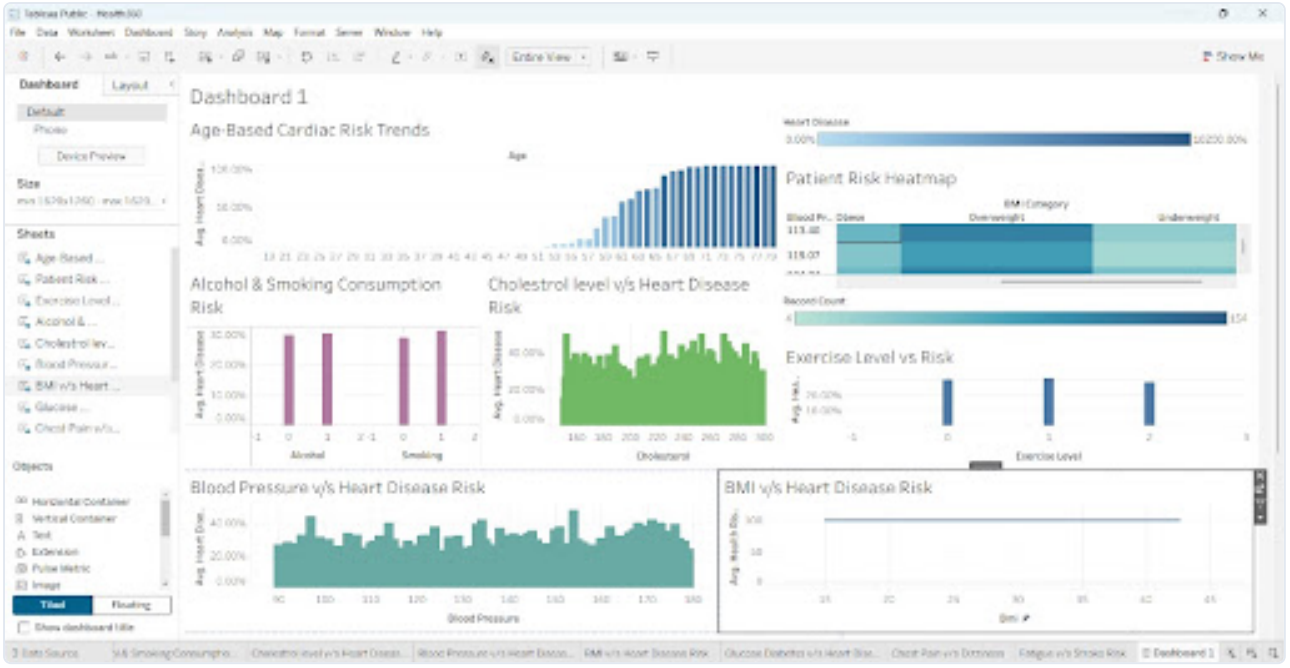
```
-- Snowflake SQL Query SELECT FATIGUE, AVG(STROKE) AS AVG_STROKE_RISK, AVG(HEART_DISEASE)
AS AVG_HEART_DISEASE FROM SMART_HEALTHCARE_DATASET GROUP BY FATIGUE;
```



Visualization 7: Fatigue v/s Stroke Risk

Q8. How can all medical metrics (Age, Risk Heatmap, BP, BMI) be integrated into a single Executive Dashboard?

```
-- Snowflake SQL View for Combined Analytics Dashboard CREATE OR REPLACE VIEW DASHBOARD_H
EALTH360_VIEW AS SELECT AGE, BMI_CATEGORY, BLOOD_PRESSURE_CATEGORY, AVG(HEART_DISEASE) AS AVG_HE
ART_RISK, AVG(HEALTH_RISK_SCORE) AS AVG_RISK_SCORE, COUNT(*) AS PATIENT_COUNT FROM SMART_HEALTHC
ARE_DATASET GROUP BY AGE, BMI_CATEGORY, BLOOD_PRESSURE_CATEGORY;
```



Visualization 8: Health360 Integrated Executive Analytics Dashboard

SUMMARY & STRATEGIC FINDINGS

- **Lifestyle Factors:** Combined alcohol consumption and smoking elevate average cardiac risk across demographic groups.
- **Biometric Trends:** Blood pressure fluctuations and high cholesterol levels directly correspond to higher health risk scores.
- **Integrated Monitoring:** The Health360 dashboard successfully bridges clinical data with visual analytics for timely health interventions.